



BITS Pilani

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Tutorial

Q1. Generate a 278×10^{-6} Hz square wave using 8253/8254.

- Given clock = 2.5 MHz
- Base address = 90H

Hints

$$\text{Counter Output freq} = \frac{\text{Input clk freq}}{\text{Count value}}$$

Count – 16-bit

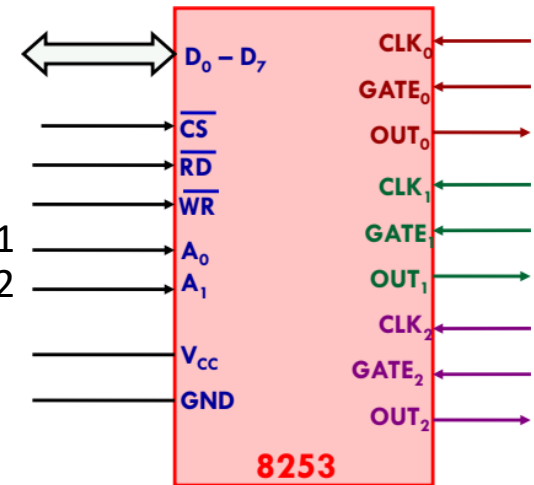
- **Required count** = $(2.5 \times 10^6) / (278 \times 10^{-6}) = 9 \times 10^9$

CS'	A ₁	A ₀	Selected
0	0	0	Counter 0
0	0	1	Counter 1
0	1	0	Counter 2
0	1	1	Control Register
1	X	X	8253/8254 Not Selected

D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀
SC1	SC0	RW1	RW0	M2	M1	M0	BCD
Selects Counter		Read/Write Control		Timer Mode			0 – binary 0000 _h FFFF _h 1 – BCD 0000 9999
00	Counter 0	00	Latch Counter	000	Interrupt on T/C		
01	Counter 1	01	R/W LSB	001	h/w re-Triggerable one shot		
10	Counter 2	10	R/W MSB	010	rate generator		
11	Read Back Command	11	R/W LSB followed by MSB	011	Square wave generator		
				1x0	s/w triggered strobe		
				1x1	h/w triggered strobe		

From
8086

A1
A2



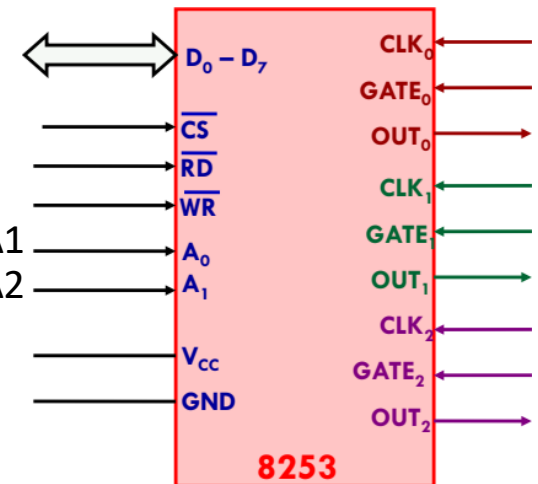
Pin Out of 8253

A7 A6 A5 A4 A3 A2 A1 A0

- Base address: 90h= 1001 0000
- Counter 0 add: 90h= 1001 0000
- Counter 1 add: 92h= 1001 0010
- Counter 2 add: 94h= 1001 0100
- CREG address: 96h= 1001 0110

CS'	A ₁	A ₀	Selected
0	0	0	Counter 0
0	0	1	Counter 1
0	1	0	Counter 2
0	1	1	Control Register
1	X	X	8253/8254 Not Selected

From
8086



Pin Out of 8253

Solution

- **Required count** = $(2.5 \times 10^6) / (278 \times 10^{-6}) = 9 \times 10^9$
- Since one counter can have max value as = 65535 ,therefore Cascading is required
- Using one more counter $65535 \times 65535 < (9 \times 10^9)$
- Hence all the counters are cascaded
- $(90 \times 10^8) = (6 \times 10^4) \times (5 \times 10^4) \times 3$
- **Control word counter-0** = 00,11,011,0 = 36H
- **Control word counter-1** = 01,11,011,0 = 76H
- **Control word counter-2** = 10,01,011,0 = 96H
- $(6 \times 10^4) = \text{EA60H}$ ---- load in counter-0
- $(5 \times 10^4) = \text{C350H}$ ---- load in counter-1
- $(3) = 3\text{H}$ ---- load in counter-2

D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀
SC1	SC0	RW1	RW0	M2	M1	M0	BCD
Selects Counter		Read/Write Control		Timer Mode			0 – binary 0000 _h FFFF _h 1 – BCD 0000 9999
00	Counter 0	00	Latch Counter	000	Interrupt on T/C		
01	Counter 1	01	R/W LSB	001	h/w re-Triggerable one shot		
10	Counter 2	10	R/W MSB	010	rate generator		
11	Read Back Command	11	R/W LSB followed by MSB	011	Square wave generator		
				1x0	s/w triggered strobe		
				1x1	h/w triggered strobe		

// **COUNTER-0**

```
MOV AL,36H
OUT 96H,AL
MOV AL,60H
OUT 90H,AL
MOV AL,EAH
OUT 90H,AL
```

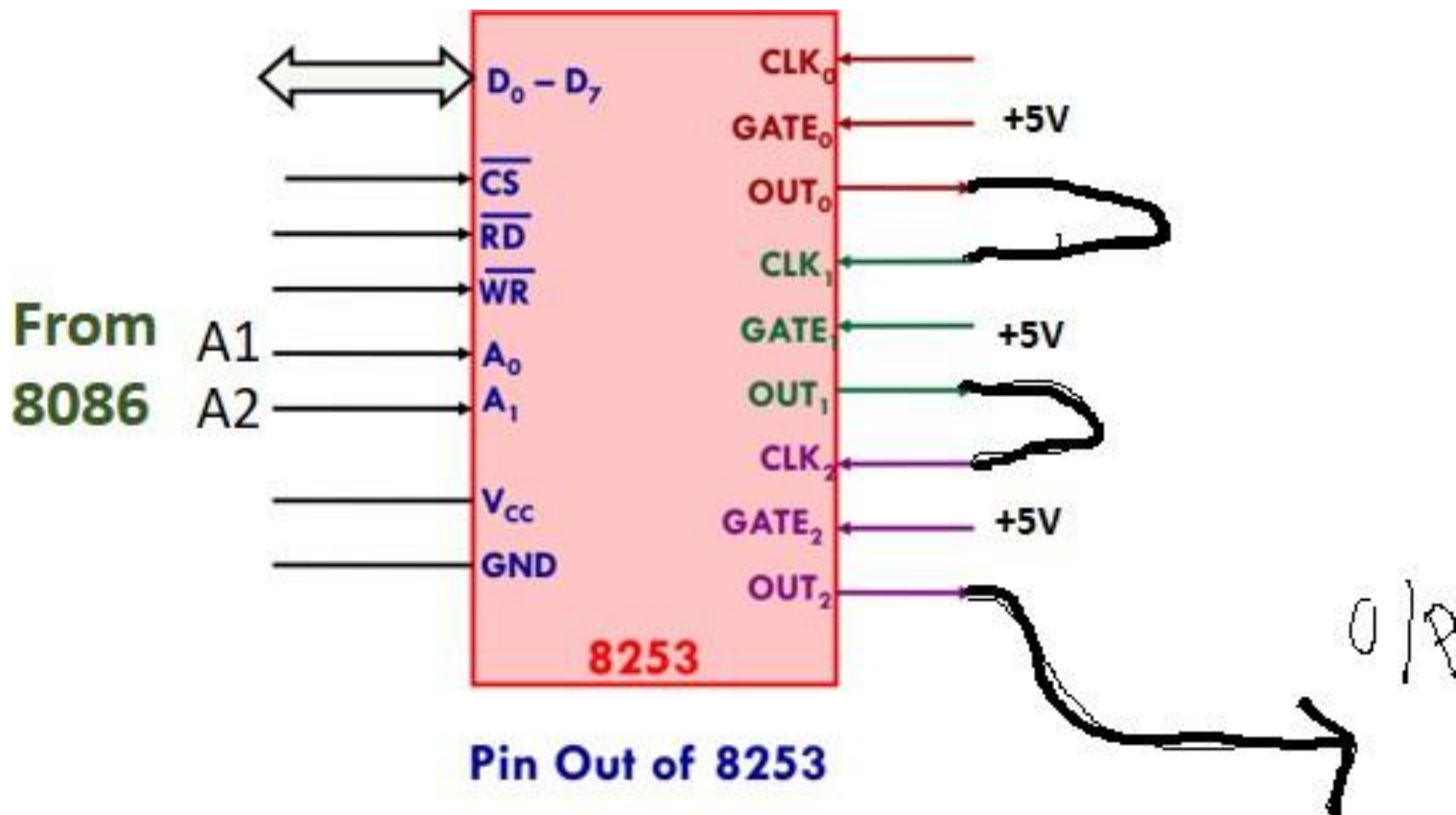
// **COUNTER-1**

```
MOV AL,76H
OUT 96H,AL
MOV AL,50H
OUT 92H,AL
MOV AL,C3H
OUT 92H,AL
```

//**COUNTER-2**

```
MOV AL,96H
OUT 96H,AL
MOV AL,03H
OUT 94H,AL
```

Counter	Address
Counter 0	90h
Counter 1	92h
Counter 2	94h
CREG	96h



Pin Out of 8253

Question 2

Initialize the 8259 for the following

Master PIC at 0x20/0x22

Slave 1 at 0xA0/0xA2 — connected to IRQ2 of the master

Slave 2 at 0xC0/0xC2 — connected to IRQ3 of the master

All operate in manual EOI mode (no AEOI)

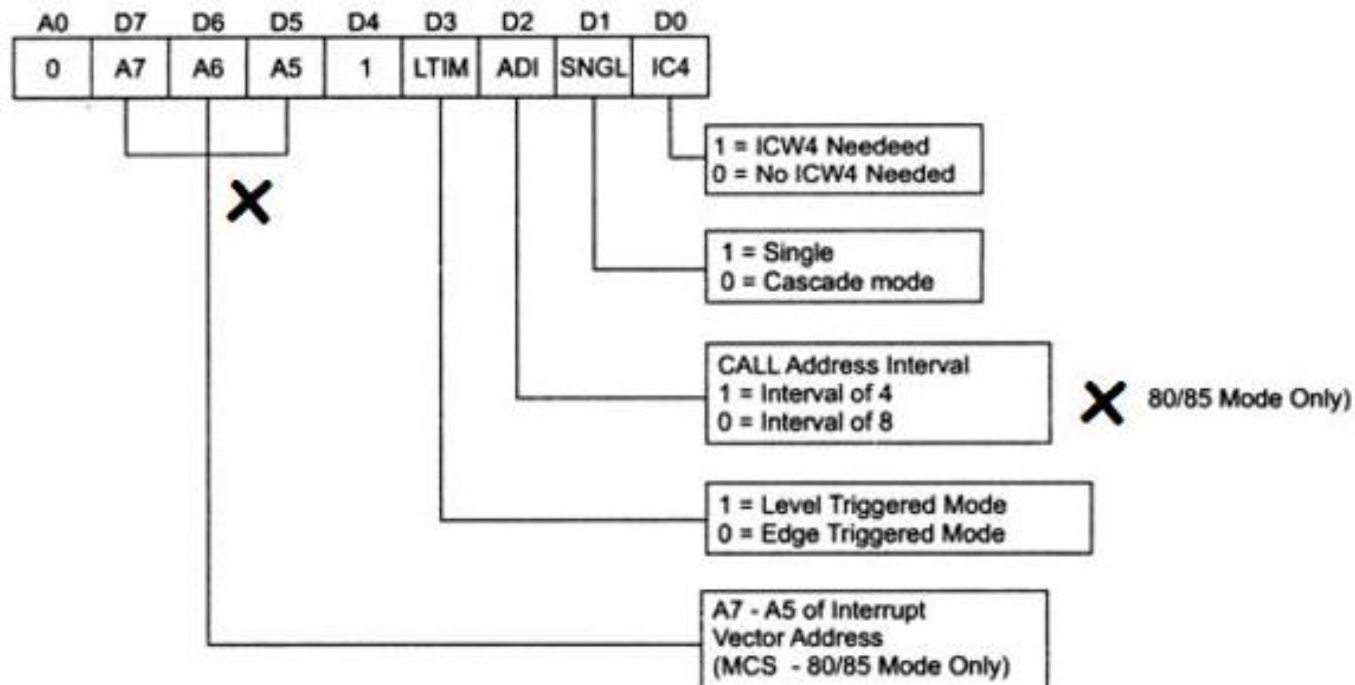
Master IRQs 0–7 => INT 0x20–0x27

Slave 1 IRQs 8–15 => INT 0x28–0x2F

Slave 2 IRQs 16–23 => INT 0x30–0x37

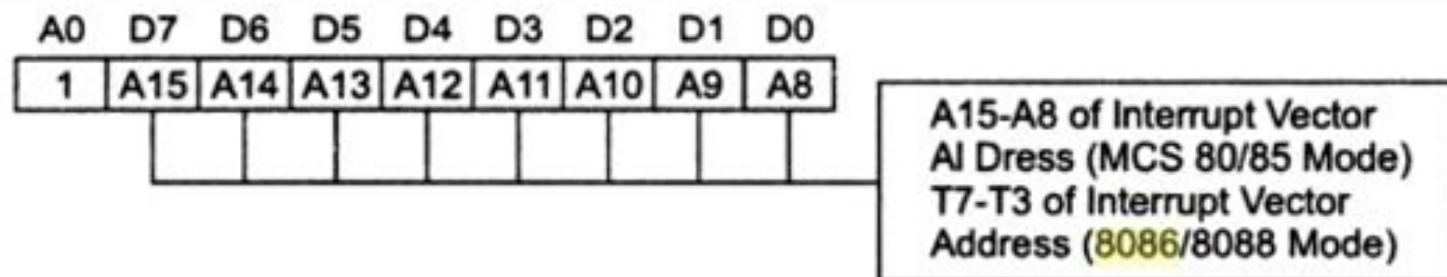
Initialization command word 1 (ICW1)

Starts the initialization sequence; sets edge/level triggering, and indicates if ICW4 is needed.



Initialization command word 2 (ICW2)

Sets the base address of the interrupt vector table (i.e., where ISRs start).

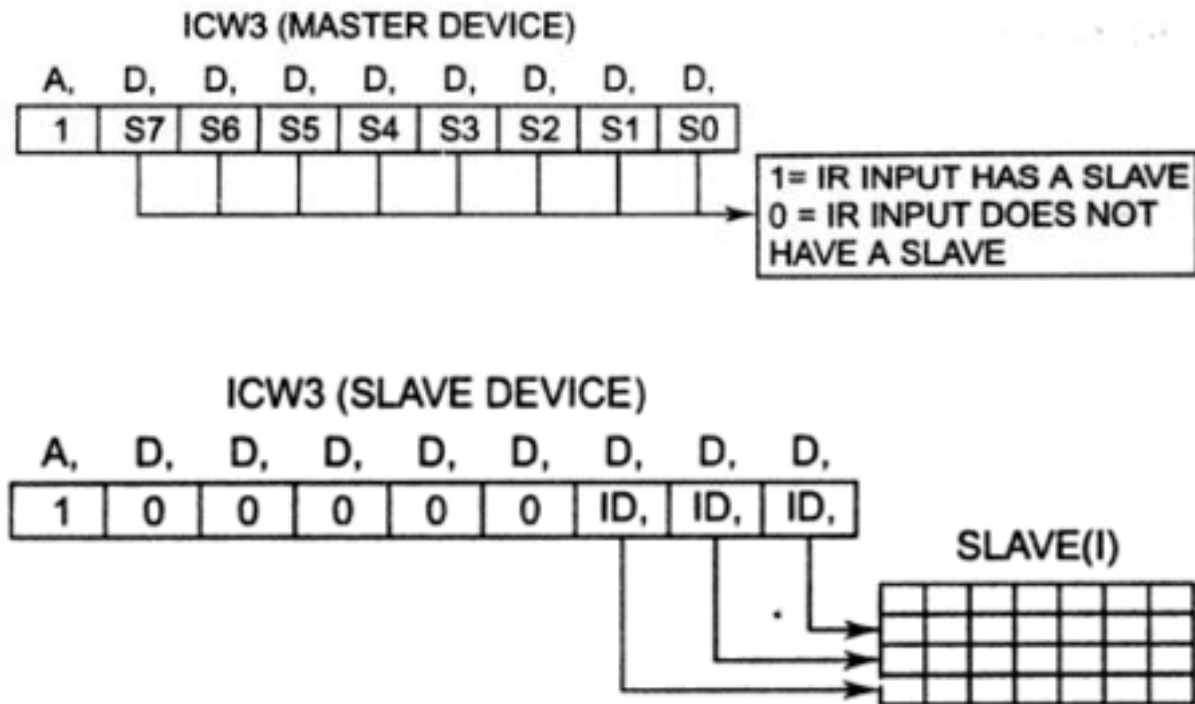


D2,D2,D0 always zero for 8086

- If you write ICW2 = 0x08 to the PIC:
- IRQ0 → INT 08h
- IRQ1 → INT 09h

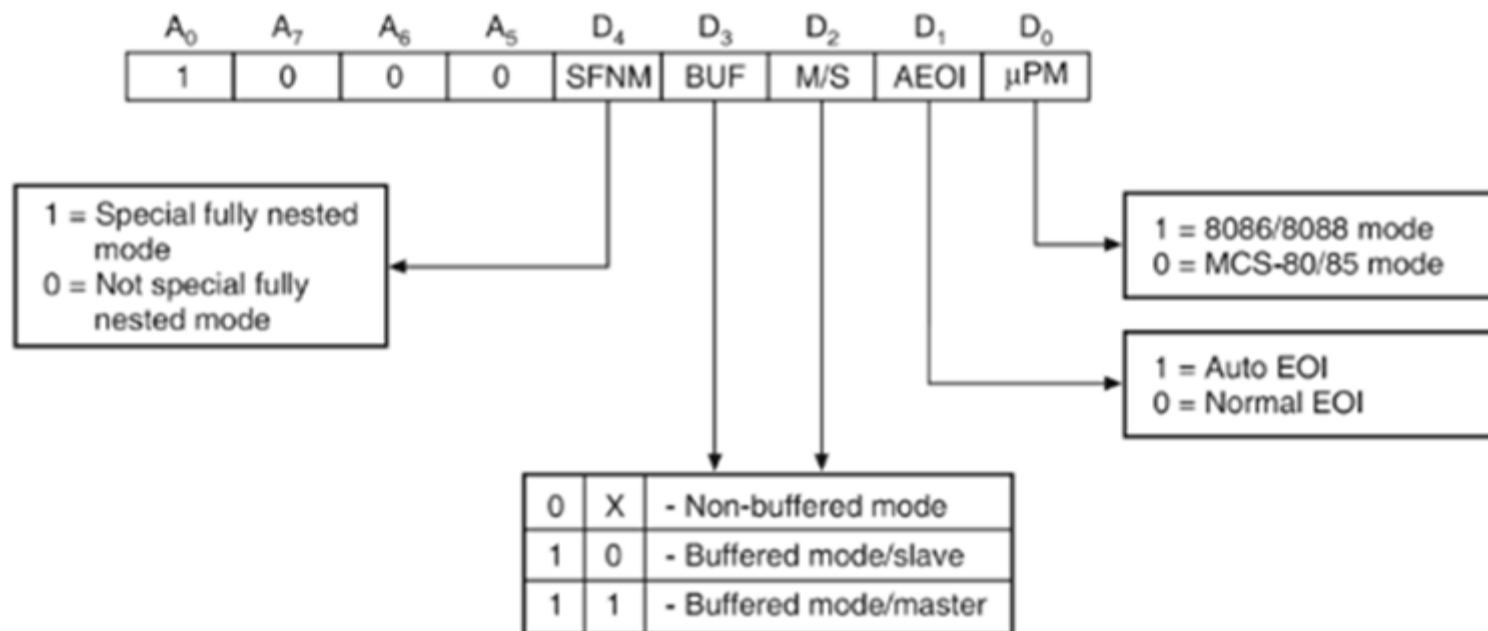
Initialization command word 3 (ICW3)

Only needed in cascaded mode



ICW3 (for slave device) indicates where the slave is connected to the master.

Initialization command word 4 (ICW4)





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ICW1: Edge triggered, expect ICW4

```
MOV AL, 0x11
OUT 0x20, AL
OUT 0xA0, AL
OUT 0xC0, AL
```

ICW2

```
MOV AL, 0x20 ;
OUT 0x22, AL
MOV AL, 0x28 ;
OUT 0xA2, AL
MOV AL, 0x30 ;
OUT 0xC2, AL
```

ICW3

```
MOV AL, 0x0C ; Master has slaves on
IRQ2 and IRQ3 (bit 2 and bit 3 set)
OUT 0x22, AL
MOV AL, 0x02 ; Slave 1 is on Master
IRQ2
OUT 0xA2, AL
MOV AL, 0x03 ; Slave 2 is on Master
IRQ3
OUT 0xC2, AL
```

ICW4: 8086/88 mode, no AEOI

```
MOV AL, 0x01
OUT 0x22, AL
OUT 0xA2, AL
OUT 0xC2, AL
```

```
;-----
; : Initial Masking
;-----
```

```
; Unmask IRQ2 and IRQ3 on Master (for both Slaves)
MOV AL, 0xF3 ; 11110011b (IRQ2 & IRQ3
unmasked)
OUT 0x22, AL
```

```
; UNMask all IRQs on both Slaves initially
MOV AL, 0x00
OUT 0xA2, AL
OUT 0xC2, AL
```